Shihua Zhang

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Biograph

I am a second-year Ph.D. student in the Department of Electrical and Computer Engineering at the National University of Singapore, advised by Prof. Xinchao Wang. Prior to joining NUS, I received my M.S. degree from Wuhan University, where I was advised by Prof. Jiayi Ma and conducted research on image matching. My current research interests include vision language models, video generation, and world models.

Education Experience

xML Lab, National University of Singapore, Singapore **2025 - present**
Ph.D. student in Department of Electrical and Computer Engineering, advised by Prof. Xinchao Wang

Multi-Spectral Vision Processing Lab, Wuhan University, Wuhan, China **2022 - 2025**
M.S. in Information and Communication Engineering, advised by Prof. Jiayi Ma

Electronic Information School, Wuhan University, Wuhan, China **2018 - 2022**
B.E. in Electronic Information Engineering

First-Author Publications

† Equal contribution.

1. **BRAVO: Bridge Matching for Autoregressive Video Generation**
   NeurIPS 2026 (Video Generation)
   Shihua Zhang†, *Zhenxiong Tan*†, *Xinchao Wang*
2. **Make Geometry Matter for Spatial Reasoning**
   ECCV 2026, **Oral, Award Candidate (0.095%)** [Paper] [Code] (VLM, Spatial Reasoning)
   Shihua Zhang, *Qiuhong Shen, Shizun Wang, Tianbo Pan, Xinchao Wang*
3. **Align Images Before You Generate**
   CVPR 2026 [Paper] [Code] (Image and Video Generation)
   Shihua Zhang, *Qiuhong Shen, Xinchao Wang*
4. **SAG-GNN: Semantic-Aware Guided GNN for Descriptor-Free 2D–3D Matching**
   CVPR 2026 [Paper] [Code] (2D–3D Matching)
   Shihua Zhang†, *Tianhao Xu*†, *Zizhuo Li, Qing Ma, Jiayi Ma*
5. **DeMatch++: Two-View Correspondence Learning via Deep Motion Field Decomposition and Respective Local-Context Aggregation**
   TPAMI 2025 [Paper] [Code] (Image Matching)
   Shihua Zhang, *Zizhuo Li, Jiayi Ma*
6. **Matching While Perceiving: Enhance Image Feature Matching with Applicable Semantic Amalgamation**
   AAAI 2025 [Paper] [Code] (Image Matching)
   Shihua Zhang†, *Zhenjie Zhu*†, *Zizhuo Li, Tao Lu, Jiayi Ma*
7. **DeMatch: Deep Decomposition of Motion Field for Two-View Correspondence Learning**
   CVPR 2024 [Paper] [Code] (Image Matching)
   Shihua Zhang, *Zizhuo Li, Yuan Gao, Jiayi Ma*
8. **DiffGlue: Diffusion-Aided Image Feature Matching**
   ACM MM 2024 [Paper] [Code] (Image Matching)
   Shihua Zhang, *Jiayi Ma*
9. **ConvMatch: Rethinking Network Design for Two-View Correspondence Learning**
   AAAI 2023, **Oral** [Paper] [Code]
   TPAMI 2024 [Paper] [Code] (Image Matching)
   Shihua Zhang, *Jiayi Ma*

Other Publications

† Equal contribution.

1. **World Action Models: A Survey**
   arXiv 2026 [Paper] (Embodied AI, World Action Model)
   *Qiuhong Shen, **Shihua Zhang**, Yue Liao, Qi Li, Zhenxiong Tan, Shizun Wang, Shuicheng Yan, Xinchao Wang*
2. **Selecting and Pruning: A Differentiable Causal Sequentialized State-Space Model for Two-View Correspondence Learning**
   TIP 2026 [Paper] (Image Matching)
   *Xiang Fang†, **Shihua Zhang**†, Hao Zhang, Xiaoguang Mei, Huabing Zhou, Jiayi Ma*
3. **CoMatch: Dynamic Covisibility-Aware Transformer for Bilateral Subpixel-Level Semi-Dense Image Matching**
   ICCV 2025, **Spotlight** [Paper] (Image Matching)
   *Zizhuo Li, Yifan Lu, Linfeng Tang, **Shihua Zhang**, Jiayi Ma*
4. **U-Match: Exploring Hierarchy-Aware Local Context for Two-View Correspondence Learning**
   TPAMI 2024 [Paper] (Image Matching)
   *Zizhuo Li, **Shihua Zhang**, Jiayi Ma*
5. **ResMatch: Residual Attention Learning for Feature Matching**
   AAAI 2024 [Paper] (Image Matching)
   *Yuxin Deng, Kaining Zhang, **Shihua Zhang**, Yansheng Li, Jiayi Ma*
6. **ACMatch: Improving Context Capture for Two-View Correspondence Learning via Adaptive Convolution**
   ISPRS 2024 [Paper] (Image Matching)
   *Xiang Fang, Yifan Lu, **Shihua Zhang**, Yining Xie, Jiayi Ma*
7. **U-Match: Two-View Correspondence Learning with Hierarchy-Aware Local Context Aggregation**
   IJCAI 2023, **Oral** [Paper] (Image Matching)
   *Zizhuo Li†, **Shihua Zhang**†, Jiayi Ma*
8. **Improving Sparse Graph Attention for Feature Matching by Informative Keypoints Exploration**
   CVIU 2023 [Paper] (Image Matching)
   *Xingyu Jiang†, **Shihua Zhang**†, Xiao-Ping Zhang, Jiayi Ma*

Internship

MT Lab, Meitu, Xiamen, China

Monocular 3D object detection *November 2021–January 2022*

• Achieve fast and accurate monocular 3D object detection and pose estimation in indoor scenes by an end-to-end network with weak shape supervision.